

Lead Architect and hands-on **Front-End / Full-Stack Engineer** with 24+ years of commercial experience, specialising in **real-time trading, risk** and operations applications built in **React, Angular, C# and TypeScript**. Builds what traders, sales, and operations staff use every day: streaming market data, high-volume blotters, RFQ and sales-to-trader workflows, trade capture, and risk and P&L dashboards. Also owns the architecture behind them, from event-driven .NET and Node services through to the front-end frameworks, SDKs and design systems that consume them.

Delivered front-office, middle-office and operations products into Millennium Management, NatWest Markets, BMO Capital Markets, ISDA, BGC, EDF Trading, and Gazprom Marketing & Trading, spanning Bonds, Repo, FX, FX Options, Swaps, Rates, Equities, Equity Derivatives, NDFs, and Futures. Led delivery of Kinetix's commercial product suite: Trade Management, Sales 360, DocHub and Contracts Intelligence, and architected the enterprise UI platforms, developer SDKs and framework infrastructure those products run on, now built on by more than ten delivery teams.

Equally at home on greenfield builds and complex existing estates, setting architecture and engineering standards, governing quality, mentoring engineers and building alongside the teams that ship. Has led modernisation across successive generations of technology: desktop to web, monolith to microservices, on-premises to cloud-native, and most recently established AI-native engineering practice on live production codebases.

CORE SKILLS & COMPETENCIES

- **Front-End Engineering:** React 18 (Hooks, Context API, Redux Toolkit), TypeScript, RxJS, Angular, Blazor, MVVM, micro-frontends, virtualised high-frequency data grids (AG Grid), Dockview, Storybook design systems, WPF, HTML5, CSS3 / SCSS, design tokens, Vite, Nx, Web Workers and rendering performance profiling.
- **Trading & Financial Technology:** Electronic trading and front-office UI, RFQ and sales-to-trader workflow, real-time pricing and market data displays, high-volume trading blotters, trade capture and lifecycle, risk and P&L dashboards, front office, middle office and pre- and post-trade workflows across Fixed Income, Bonds, Repo, Equities, Equity Derivatives, FX, FX Options, Futures, NDF, Swaps, Rates, Allocations and Compliance (MiFID II).
- **Distributed & Real-Time Systems:** Microservices, event-driven architecture, RxJS and Rx reactive streams, Kafka, RabbitMQ, Azure Service Bus, WebSockets, SignalR, Protobuf, MessagePack, low-latency and highly available systems.
- **Financial Desktop Platforms:** OpenFin, Glue42, Interop.io, FDC3, workspace integration, workflow automation and cross-application communication.
- **Data & Analytics Platforms:** Market data, real-time data visualisation, geospatial mapping and operational dashboards; Syncfusion, Kendo UI, Highcharts and MapLibre GL JS.
- **Backend, .NET & Desktop Engineering:** C#, .NET 8, ASP.NET Core, Web APIs, GraphQL, Microsoft Graph API, Node.js, Entity Framework, LINQ, Rx.NET, TPL Dataflow, asynchronous programming, Python; WPF, Prism, MVVM, Excel-DNA and Win32 interop for hybrid web/desktop delivery.
- **Architecture & Platform Engineering:** Enterprise and solution architecture, domain-driven design, enterprise UI platforms, developer SDKs, micro-frontends, shared services, design systems, API design, performance engineering, SaaS delivery, application scaffolding, technology modernisation and identity & access management (OAuth2/OIDC, SSO, JWT, Keycloak).
- **Quality & Test Engineering:** Vitest, Jest, Playwright, Cypress, Storybook, NUnit, xUnit, SpecFlow, component and visual regression testing, UI automation, TDD, BDD and DDD.
- **Cloud, Data & Delivery:** AWS, Azure, Docker, Kubernetes, Helm, ArgoCD, Terraform, Git, GitHub Actions, Azure DevOps, CI/CD, SQL Server, PostgreSQL, Oracle, MongoDB, Redis, DuckDB and Elasticsearch.
- **AI-Native Engineering & Technical Leadership:** AI-assisted development, LLM context engineering, multi-agent development workflows, Claude Code and Cursor; architecture governance, technical strategy, roadmaps, mentoring, code review and agile delivery across distributed teams.
- **Industry Expertise:** Hedge funds, capital markets, commodities and energy trading, healthcare, retail and e-commerce.

EXPERIENCE

Lead Architect / Principal Front-End Engineer | Kinetix Trading Solutions

Oct 2025 - Sep 2026

Engagement: a global multi-strategy hedge fund | Fixed Income Transactional Systems and Commodity Trading

Lead architect and hands-on React / Angular / TypeScript engineer across a programme of front-office, back-office and operations applications delivered within a Fixed Income Transactional Systems and Commodity Trading group. Built the applications alongside the desks consuming them, while owning the enterprise platform and SDK that made delivery scalable across teams.

Applications Delivered

- **Risk Tech Monitor** - real-time risk monitoring and trading-desk operations dashboard aggregating multi-source risk signals, live positions and P&L into a single operational view, with configurable alerting and drill-down analytics.
- **Bridge IT** - cross-application interoperability platform on OpenFin and FDC3. Carries context, events and workflow automation between otherwise unconnected trading applications, turning a fragmented desktop into a single trader workbench.
- **Post-Trade P&L Sign-Off Workflow** - end-to-end trade P&L review, exception management and multi-stage approval workflow spanning front and middle office, with full audit trail, compliance controls and integration into downstream reporting systems.
- **Vessel Tracking** - real-time geospatial platform for commodity and shipping operations: live fleet positions, route visualisation and operational status dashboards, built on MapLibre GL JS with RxJS streaming pipelines over WebSockets.

Platform, SDK & Engineering

- Engineered the RxJS streaming pipelines carrying live pricing, market and operational data: multicast subjects, backpressure handling, throttling, windowing and conflation. High-frequency updates land in virtualised blotters without the UI stalling.
- Optimised application performance for high-frequency data environments through virtualised rendering, memoisation, selective re-render strategies, Web Worker offloading and bundle-level code splitting.
- Designed and delivered an enterprise UI platform of reusable data grids, geospatial visualisation, charting, docking workspaces and workflow components, configured through metadata rather than rebuilt per application.
- Architected a framework-agnostic SDK with a zero-dependency core, so business logic, services and workflows run unchanged under React, Angular, mobile and Node. More than ten delivery teams build on it today.
- Established the real-time streaming architecture: live data distribution over WebSockets, claims-based authorisation, user-targeted messaging and horizontally scalable Kubernetes deployments.
- Established a unified enterprise design system and component architecture, documented in Storybook, so applications written by different teams and on different stacks look and behave the same.
- Embedded quality engineering across the platform: Vitest unit and component tests, Playwright end-to-end automation and visual regression coverage, with one set of test standards for every consuming team.
- Wrote the CLI scaffolding framework that generates a working React or Angular application with authentication, containerisation, CI/CD, Kubernetes manifests and AI context already wired in.
- Built the AI-native developer platform: structured context management and package-level documentation, so Claude Code and Cursor generate framework-compliant code without weeks of onboarding.
- Introduced a multi-agent engineering model of role-based AI agents, persistent context and controlled handoff, running alongside developers on active production codebases.
- Led a team of three, owning coding standards, code review practices and architectural governance across everything built on the platform.

Technology Stack: React 18, Angular 22, TypeScript, RxJS, .NET 10, Redux Toolkit, Context API, MVVM, NgRx, AG Grid, Highcharts, MapLibre GL JS, Dockview, Ant Design, NG-ZORRO, Figma, WebSockets, Socket.IO, Vite, Nx, SCSS, CSS Design Tokens, HTML5, Storybook, Vitest, Jest, Playwright, JWT, Docker, Kubernetes, ArgoCD, AWS, GitHub Actions, Claude Code, Python, Cursor, Multi-Agent AI, LLM Context Engineering.

Architect and hands-on technical lead across Kinetix commercial product suite: Trade Management, Sales 360, DocHub and Contracts Intelligence. All four run on the Kinetix enterprise platform, a React / TypeScript UI framework, .NET service layer and reusable SDK. Products spanned Bonds, Repo, FX, FX Options, Equities, Equity Derivatives, Futures, NDF, Swaps, Rates, Allocations and Compliance across pre- and post-trade workflows, delivered to Tier-1 banks, brokers and industry bodies as white-labelled and SaaS solutions. Responsible for platform architecture, interoperability, developer tooling, cloud integration and engineering standards across web and desktop.

Products Delivered

- **Trade Management** - cross-asset trade capture across electronic and voice execution, position management, post-trade event processing, confirmations, allocations, exception management, straight-through downstream integration and MiFID II compliance, covering Bonds, Repo, FX, FX Options, Equity Derivatives, Swaps, Rates, NDFs and Futures. Its largest deployment retired a 25-year-old bond and futures system across US and Asian desks, took on 20+ straight-through processing feeds including Bloomberg TOMS, and cut annual operating expenses by \$5.3M and maintenance costs by 47%.
- **Sales 360** - consolidated client, coverage and revenue platform for a North American investment bank's sales and trading business, covering RFQ and sales-to-trader workflow, client engagement analytics, pipeline visibility and cross-desk operational reporting.
- **DocHub (Digital Document Library)** - white-labelled standards and rulebook platform for a global derivatives industry body: interactive rendering of digitised documents, version comparison and blacklining, AI-assisted search, relationship mapping across definitions and supplements, personalisation and annotation, and API-based distribution to member firms. Cut the body's document digitisation timelines 60-fold.
- **Contracts Intelligence** - AI-driven analytics over ISDA, GMRA / GMSLA, CSA, prime broker and FCM agreements: cross-corpus agentic search, clause and term standardisation, full versioning and amendment tracking, one-click playbook alignment checks and portfolio-wide risk and exposure analysis.

Platform & Engineering

- Architected and delivered the React / TypeScript MVVM framework behind the suite: application lifecycle management, dependency injection, plugin architecture, SDK distribution and desktop interoperability across OpenFin and Interop.io.
- Built the trading-platform APIs delivery teams used to ship line-of-business features across equity derivatives and FX, into React and Blazor web front ends and OpenFin and Glue42 desktop containers.
- Designed and built C# / .NET 8 and Blazor services, and defined the API contracts and integration standards for the cloud-native .NET and Java microservices behind them, built on Kafka, RabbitMQ and Elasticsearch.
- Designed a loosely coupled micro-frontend architecture with RxJS event-driven communication, runtime service composition, permission-based feature delivery and application-scoped event aggregation.
- Built reactive business rules, command processing and a forms framework on RxJS observable streams, underpinning high-volume trading workflow.
- Delivered data virtualisation and metadata-driven configuration so high-volume trading blotters render and update in real time without client-side degradation.
- Integrated OpenFin, Interop.io and FDC3 so applications from different vendors share context and hand work to each other on the trader desktop.
- Set up a platform-engineering model in which delivery teams composed applications from reusable frameworks, shared services, developer SDKs and common UX patterns instead of rebuilding each one from scratch.
- Built voice-driven workflows, conversational interfaces and intent-based automation into the product's real-time event-driven architecture.
- Established CI/CD pipelines and the Playwright and Cypress automation suites, with TDD / BDD practice across distributed engineering teams.
- Led a team of four developers across the product suite, owning technical direction, code review and delivery standards, and worked directly with client engineering teams through installation and integration.

Technology Stack: React, TypeScript, RxJS, MVVM, C#, .NET 8, Python, Blazor, Node.js, Microsoft Graph API, OpenFin, Glue42, Interop.io, FDC3, AG Grid, Kendo UI, PrimeNG, ProseMirror, WebSockets, Web Workers, Kafka, RabbitMQ, ActiveMQ, Elasticsearch, Kibana, Redis, DuckDB, SQL Server, MongoDB, Docker, Kubernetes, AWS, Azure, Terraform, Cypress, Jest, Azure DevOps, CI/CD, TDD, BDD, Agile.

Senior Consultant / Front-End & Platform Engineer | Kinetix Trading Solutions Jun 2017 - Feb 2020

Engagement: a Tier-1 investment bank | electronic Repo and Bond trading platform

Responsible for architecture and development of client applications, core framework components and backend services for a greenfield, high-availability, low-latency multi-asset core banking and trading platform used by major financial institutions across the UK and US markets.

Key Achievements

- Delivered Repo and Bond electronic trading applications across desktop and web - trade capture, order and execution workflow, market data displays and real-time trading blotters used by trading and sales desks.
- Built the core platform capabilities other delivery teams used to ship trading functionality across desktop and web.
- Built the reactive data and event-handling layers on Reactive Extensions (Rx) and event aggregation. Market data and order-state changes reached the blotters with minimal latency.
- Designed and delivered the data virtualisation, configuration, command and forms frameworks used across the trading applications.
- Architected a loosely coupled, SDK-based application model for modular development, extensibility and simpler client integration.
- Defined API specifications and integration patterns between client applications and Java-based backend microservices.
- Established automated testing, CI/CD pipelines, UI automation frameworks and engineering governance practices.

Technology Stack: C#, WPF, MVVM, React, TypeScript, Reactive Extensions (Rx), microservices, LINQ, WebSockets, Event Aggregation, Telerik, DevExpress, Autofac, TPL Dataflow, Multithreading, Docker, Kubernetes, AWS, Azure, ClickOnce, CI/CD, TeamCity, Jenkins, PowerShell, TDD, BDD, SpecFlow, SQL Server, Oracle, Agile.

EARLIER EXPERIENCE (2002 - 2017)

Full-Stack Developer | EDF Trading

Feb 2016 - Apr 2017

Full-stack delivery for one of Europe's largest energy trading businesses: Angular front ends, WPF desktop applications, .NET backend services and Excel-DNA integrations, with real-time SignalR data visualisation. Built the shared frameworks and enterprise NuGet packages adopted across teams, and the CI/CD pipelines on TFS Release Management. Technologies: C#, .NET 4.6, Angular, TypeScript, WPF, MVVM, WCF, Web API, Rx, SignalR, Azure Service Bus, SQL Server, Oracle, Telerik, DevExpress, Excel-DNA, TDD / BDD.

Senior .NET Consultant | Graphnet Health

Sep 2014 - Dec 2015

Architected Service Dashboard, a real-time monitoring and diagnostics platform handling millions of service-bus messages with live visualisation of data flows, and Data Integrator, a graphical workflow studio that let product specialists transform heterogeneous data sources and publish them through REST APIs. Technologies: C#, .NET 4.5, WPF, Prism, MVVM, Rx, AngularJS, TypeScript, Web API, SignalR, NServiceBus, TPL Dataflow, Autofac, RavenDB.

Senior .NET Developer | Gazprom Marketing & Trading

Apr 2014 - Aug 2014

Delivered architecture and reusable framework components for Power and Gas storage trading across UK and European markets, integrating OpenLink Endur (CTRM), Excel-based workflows and internal enterprise services. Technologies: C#, .NET, WPF, MVVM, Prism, Rx, Web API, SignalR, RabbitMQ.

Senior .NET Consultant | DRS (Data Research and Services)

May 2013 - Apr 2014

Built the MVVM application, navigation, persistence, localisation, activity-tracking and weak-event frameworks used across a global e-marketing product suite, with high-performance document controls and early HTML5 / Angular SPA proofs-of-concept that set the product's future direction. Technologies: C#, WPF, MVVM, Prism, MEF, Rx, ASP.NET MVC, AngularJS, HTML5, Web API, Telerik, NUnit.

Senior WPF Developer | UST Global

Oct 2012 - Apr 2013

For a healthcare software provider, delivered reusable APIs and framework components, desktop capabilities including hotkey navigation and layout personalisation, and support-assistant tooling adopted across multiple healthcare products by development, QA and operations teams. Technologies: C#, WPF, MVVM, Prism, Castle Windsor, WCF, Win32, SQL Server, DevExpress, NUnit.

Technical Architect | Steria / Xansa

Aug 2004 - Sep 2012

For a major UK retail and healthcare group, architected large-scale healthcare and retail platforms including clinical workflow systems and operational dashboards, built reusable WPF and web frameworks adopted across projects, and led delivery teams and major transformation initiatives. Technologies: C#, ASP.NET MVC, WPF, Prism, Silverlight, WCF, SQL Server, Oracle, MongoDB, NServiceBus, Entity Framework, ServiceNow, L1/L2 Support.

Developer | Mindmill Software Ltd

Jun 2002 - Jul 2004

Delivered banking, insurance, aviation and public-sector solutions for international clients, including flight operations information system functionality, flight data processing components and biometric attendance systems for a national research organisation. Technologies: ASP.NET, JavaScript, HTML / CSS, SQL Server, MySQL, XML / XSLT, VB6, VC++, COM.

EDUCATION & ADDITIONAL INFORMATION

Education

- **Master of Information Technology** (MIT Computers)
- **Bachelor of Science** (Electronics).

Additional Information

- **British Citizen**, based in **London**, UK. **Author of 100+ technical articles** and **open-source contributor**.
- Interests: Technology R&D, Electronics and Circuit Design.

CERTIFICATIONS

- React, JavaScript, CSS.
- C#, Python, Node.js, REST API, SQL.
- Problem Solving, IKM Assessment, ITIL Foundation.
- All details: <https://pixytech.com/certifications>